

KIMBERLY K. BROUGHTON

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Software Engineer with over 17 years of experience designing, developing, and modernizing enterprise software solutions. Proven expertise in Python, JavaScript, automation, APIs, and database design across scientific research, healthcare, publishing, and government environments. Recognized NASA Inventor with a history of delivering scalable applications, reducing technical debt, automating complex workflows, and collaborating directly with stakeholders to transform business requirements into reliable software solutions.

SIGNIFICANT ACHIEVEMENTS

Sole architect and lead developer of NASA Earthdata Pub Overview and Forms applications used by all 12 NASA DAACs and all Data Producers. Official NASA “Inventor” of Earthdata Pub (NASA Case No. GSC-18,612-1).

Automated publishing workflows from hours of manual work to a single button click, virtually eliminating errors and improving costs (02/2018).

Contributed to ONC Stage 1 and Stage 2 Electronic Health Record certification as one of three assigned developers.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, GraphQL, C#, C++, Java, HTML5, CSS3, XML, Shell, AWK

Frameworks: Flask, Vue.js, React, SQLAlchemy, Angular, Bootstrap, jQuery, Jinja2, Cypress

Cloud & DevOps: AWS Cloud, Docker, Kubernetes, Git, GitLab, GitLab CI/CD Pipelines, GitLab Runners, YAML, Automation Scripting

Databases: PostgreSQL, Oracle, Microsoft SQL Server, MySQL, Informix

Development Tools: VS Code, DBVisualizer, Visual Studio, Swagger, SpectaQL, Sphinx, Vuesse

Content Management: Drupal

AI Development Tools: Cline, DUO AI, GitHub Copilot

Accessibility: Accessibility-minded front-end development aligned with WCAG AA principles

Operating Systems: Linux, Microsoft Windows

CORE COMPETENCIES

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|---------------------------|------------------------|-----------------------------|
| • Software Architecture | • Database Design | • Application Modernization |
| • Full Stack Development | • Automation | • Technical Documentation |
| • Enterprise Applications | • System Integration | • User Experience |
| • REST APIs | • Technical Leadership | • Agile Development |
| • Authentication | | |

EXPERIENCE

NASA / ORNL DAAC – Oak Ridge, TN

04/2018 – Present

Systems Architect / Application Developer

Developed in: Python, JavaScript (above frameworks and DevOps), SQLAlchemy, GraphQL, AWS Cloud, Perl, HTML, CSS, SQL, Drupal, DBVisualizer, VS Code

- Repurposed NASA's Cumulus Dashboard as the [NASA Earthdata Pub Dashboard](#) powering [Overview](#) and [Forms](#) applications (Vue, React) for all 12 NASA DAACs and all Data Producers
- Co-developed the Earthdata Pub REST API powering multiple NASA applications
- Assisted in refactoring and expanding River Function Indicator Questionnaire (RFIQ) to a generalized [Environmental Decision Support Toolkit](#)
- Designed and implemented MyWorkSpace, a modular data-order application expanding user download options
- Delivered an Ingest Dashboard with a three-developer team for ORNL DAAC ingest operations (MVP live), architected to interface with Earthdata Pub for QA and full automation
- Automated processing workflows, reducing manual processing time by 80%
- Modernized sites and scripts from Perl to Python using an efficient templating system
- Implemented automated unit and end-to-end testing to improve deployment confidence and reduce regression defects
- Built metadata import/export pipelines supporting multiple formats and standards

- Modernized and maintained the ORNL DAAC web platform while supporting migration to NASA's cloud-hosted Earthdata environment, improving maintainability and supporting future platform growth

COSBY HARRISON COMPANY (Formerly TAP Publishing Company) – Crossville, TN 10/2016 – 04/2018

System Architect / Web Application Developer / Web Developer

Developed in: Python, JavaScript, Perl, DOS, AutoHotKey, HTML, CSS, SQL, Oracle, Informix, Shell, C++

- Automated publishing workflows from hours of manual work to a single button click, virtually eliminating errors and improving costs
- Standardized data across multiple formats through automated import/export pipelines
- Automated tasks via scripts across multiple environments and languages
- Maintained and enhanced custom, publication, and company web platforms with feature and framework improvements

LIBERA, INC. – Knoxville, TN

10/2015 – 07/2016

Web Application Developer

Developed in: VB.NET / ASP.NET, XML, C#, TSQL, System7, SQL, Visual Studio 2015

- Implemented Vocational Rehabilitation program updates affecting all 78 federally funded state agencies.
- Created custom queries and scripts enabling agency end users to access timely reporting data
- Performed unit testing for statewide Vocational Rehabilitation program updates supporting developers across the state
- Wrote operating instructions and reference documentation for end users

RMB, INC. – Knoxville, TN

03/2015 – 07/2015

FACS Application Programmer

Developed in: FACS, VBScript, AutoHotKey, Jasper Reports, Crystal Reports

- Maintained database operations at peak performance, protecting secured information and ensuring data integrity
- Built daily jobs and import/export routines routing high-volume data in and out of production systems
- Created custom queries and reports providing timely and accurate information to end users
- Wrote operating instructions and reference documentation for users

QRS, INC. – Knoxville, TN

05/2009 – 02/2015

EHR Developer / Software Developer

Developed in: JavaScript, HTML, CSS, PHP, Python, Perl, Linux, MySQL, MSSQL, c-tree

- Indirectly led other EHR programmers on client-facing clinical software projects
- Contributed to ONC Stage 1 and Stage 2 Electronic Health Record certification as one of three assigned developers.
- Designed Standard Form Sets / Notes tailored to doctor specialty
- Worked directly with doctors to determine practice needs and translate them into software solutions
- Provided Tier 2 product support, resolving escalated client issues
- Created documentation and instructional videos training clients on Meaningful Use compliance and adoption
- Implemented HL7 interfaces and data conversions based on client needs
- Designed back-end and front-end interfaces and custom programs based on client needs

CERTIFICATIONS & ASSESSMENTS

UI / UX Certification – NN/g Nielsen Norman Group

12/2020

C# Certification – Knoxville, TN

12/2017

JavaScript Assessment – [SHL](#) – Expert (4.51/5.00)

08/2015

- Session ID: 818485129153777500

EDUCATION

PELLISSIPPI TECHNICAL COMMUNITY COLLEGE – Knoxville, TN

05/2007 – 05/2009

- *A.A.S. Computer Programming*
- *A.A.S. Database Design and Development*
- GPA: 3.60
- Dean's List for Fall 2008 and Spring 2007